

1. Administrative & Study Identification

Full Study Title: A Phase 3 Randomized, Active-controlled, Open-label, Multicenter Study to Compare the Efficacy and Safety of MK-2870 Monotherapy Versus Treatment of Physician's Choice as Second-line Treatment for Participants with Recurrent or Metastatic Cervical Cancer **Short Title/Acronym:** TroFuse-020 / GOG-3101 / ENGOT-cx20 **Protocol Number:** MK-2870-020 **NCT Number:** NCT06459180 **Sponsor Name:** Merck Sharp & Dohme LLC **Investigational Product:** Sacituzumab Tirumotecan (MK-2870) **Phase of Development:** Phase III **Trial Status:** Recruiting **Medical Monitor:** Merck Clinical Operations Lead [Name and Contact Information]

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: This study consists of two phases: a sacituzumab tirumotecan safety run-in and a Phase 3 portion. The primary study hypotheses are that, in the Phase 3 portion, sacituzumab tirumotecan results in a superior overall survival (OS) compared to treatment of physician's choice (TPC) in participants with high trophoblast cell surface antigen 2 (TROP2) expression levels and in all participants. **Secondary Objectives:** To evaluate the efficacy and safety of sacituzumab tirumotecan at the dose for evaluation in the Phase 3 portion, including safety, tolerability, Progression-Free Survival (PFS), Objective Response Rate (ORR), and Duration of Response (DOR).

2.2 Background & Rationale Recurrent or metastatic cervical cancer has a poor prognosis, especially after progression on first-line platinum-based chemotherapy and anti-PD-1/anti-PD-L1 therapy. There is a high unmet medical need for effective second-line treatments. MK-2870 is an investigational antibody-drug conjugate (ADC) targeting the TROP2 antigen, designed to deliver a cytotoxic payload (KL610023) directly to cancer cells to destroy them and improve survival outcomes.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Treatment of Physician's Choice / TPC) **Blinding:** Open-label **Randomization:** Randomized (1:1 Ratio in Phase 3)

3.2 Planned Enrollment & Duration Target \$N\$: Approximately 686 participants **Number of Sites:** Multicenter (~152 locations across

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Assigned female sex at birth, at least 18 years of age at the time of providing informed consent. 2. Histologically-confirmed diagnosis of squamous cell carcinoma, adenosquamous carcinoma, or adenocarcinoma of the cervix. 3. Must have recurrent or metastatic cervical cancer that has progressed on or after treatment with 1 prior line of systemic platinum doublet chemotherapy (with or without bevacizumab) AND must have received anti-PD-1/anti-PD-L1 therapy as part of prior cervical cancer regimens. 4. Has measurable disease per Response Evaluation Criteria In Solid Tumors (RECIST) 1.1, as assessed by the investigator. (Lesions situated in a previously irradiated area are considered measurable if progression has been shown). 5. Has ECOG performance status of 0 or 1 within 7 days before allocation for the run-in or within 7 days before randomization for the Phase 3 portion. 6. Has provided tumor tissue (most recent sample is preferred) from a core or excisional biopsy of a tumor lesion not previously irradiated. 7. HIV-infected participants must have well-controlled HIV on antiretroviral therapy (ART). 8. Hepatitis B surface antigen (HBsAg) positive participants are eligible if receiving HBV antiviral therapy for ≥ 4 weeks with an undetectable viral load; history of Hepatitis C virus (HCV) infection is eligible if HCV viral load is undetectable at screening. 9. Has adequate organ function.

4.2 Exclusion Criteria 1. Has Grade ≥ 2 peripheral neuropathy. 2. Has history of documented severe dry eye syndrome, severe Meibomian gland disease and/or blepharitis, or severe corneal disease that prevents/delays corneal healing. 3. Has active inflammatory bowel disease requiring immunosuppressive medication or previous history of inflammatory bowel disease (e.g., Crohn's disease, ulcerative colitis, or chronic diarrhea). 4. Has uncontrolled, significant cardiovascular disease or cerebrovascular disease. 5. Received prior systemic anticancer therapy outside of allowed lines. 6. Received prior radiotherapy within 2 weeks of start of study intervention, or has radiation-related toxicities requiring corticosteroids. 7. Received a live or live-attenuated vaccine within 30 days before the first dose of study intervention. 8. Has other histological subtypes of cervical cancer (e.g., carcinosarcoma) or nonepithelial cancer (e.g., sarcoma, neuroendocrine tumors) of the cervix. 9. Known additional malignancy that is progressing or has required active treatment within the past 3 years. 10. Known active central nervous system (CNS) metastases and/or carcinomatous meningitis. 11. Active infection requiring systemic therapy. 12. HIV-infected participants with a history of Kaposi's sarcoma and/or Multicentric Castleman's Disease. 13. Concurrent active Hepatitis B and active Hepatitis C virus infection. 14. Severe

hypersensitivity ($\geq$ Grade 3) to sacituzumab tirumotecan or treatment of physician's choice (TPC) and/or any of their excipients, or other biologic therapy. 15. Participants who have not adequately recovered from major surgery or have ongoing surgical complications. 16. Has a history of (noninfectious) pneumonitis/ILD that required steroids or has current pneumonitis/ILD. 17. Prior treatment with an ADC similar to MK-2870. 18. Pregnant or breastfeeding females.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: - *Experimental Arm:* Sacituzumab Tirumotecan (MK-2870) administered once every 2 weeks. - *Active Comparator Arm (TPC):* Pemetrexed, Topotecan, Vinorelbine, Gemcitabine, Irinotecan, or Tisotumab Vedotin administered at protocol-specified intervals (every 3 or 6 weeks). **Route of Administration:** Intravenous (IV) infusion **Duration of Treatment:** Continued until documented disease progression, unacceptable toxicity, or study withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care medications to manage side effects (e.g., antiemetics), administration of killed vaccines. **Prohibited:** Concurrent investigational agents, live/live-attenuated vaccines within 30 days before the first dose, or other systemic anti-cancer therapies outside the protocol.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Tissue Collection for TROP2 testing, Baseline Imaging, Viral Load Testing (HIV/HBV/HCV)
Baseline	Day 0	Randomization, First Dose Administration, ECOG assessment
Interim	Every 2, 3, or 6 Weeks	Treatment Administration, Safety Labs, AE Monitoring, Physical Examinations
End of Study	Disease Progression	Final Safety Assessment, Tumor Imaging, Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints Phase 3 Portion Hypothesis Endpoint: 1. **Overall Survival (OS):** evaluated in participants with high TROP2 expression and in all participants.

Sacituzumab Tirumotecan Run-in Endpoints: 1. **Objective Response Rate (ORR):** Percentage of participants with Complete Response (CR) or Partial Response (PR) per RECIST 1.1 as assessed by Blinded Independent Central Review (BICR). 2. **Number of Participants Experiencing One or More Adverse Events (AEs).** 3.

Number of Participants Discontinuing Study Treatment Due to an AE.

7.2 Secondary Endpoints (Phase 3 Portion) 1. **Progression-Free Survival (PFS)** 2. **Objective Response Rate (ORR)** 3. **Duration of Response (DOR)**

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous collection at each study visit (IV infusion appointments) through blood tests, physical exams, and patient questionnaires. An AE is defined as any untoward medical occurrence temporally associated with the use of study intervention. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Independent committee established to evaluate safety and efficacy during the initial sacituzumab tirumotecan safety run-in and the broader Phase 3 portion.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for efficacy outcomes; Safety Set for all AE evaluations. **Sample Size Justification:** Powered to detect a statistically significant superiority in Overall Survival (OS) for the MK-2870 arm compared to TPC. **Handling of Missing Data:** Kaplan-Meier methodology for time-to-event data (OS, PFS, DOR) with appropriate censoring for patients lost to follow-up.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring of trial sites to ensure data integrity and protocol adherence. **Audit Trail:** Electronic Data Capture (EDC) system tracking, centralized pathology review for TROP2 expression, and independent centralized imaging review (BICR).

11. Study Identifiers & Governance

Primary ID: 2870-020 **Registry Number:** NCT06459180 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** MK-2870 in Second-line Recurrent or Metastatic Cervical Cancer **Official Regulatory Title:** A Phase 3 Randomized, Active-controlled, Open-label, Multicenter Study to Compare the Efficacy and Safety of MK-2870 Monotherapy Versus Treatment of Physician's Choice as

12. Clinical Framework (Cervical Cancer Specific)

Primary Condition: Recurrent or Metastatic Cervical Cancer
Diagnosis Threshold: Disease progression on or after 1 prior line of platinum doublet chemotherapy and anti-PD-1/anti-PD-L1 therapy
Medical Methodology: TROP2-directed Antibody-Drug Conjugate (ADC) therapy
Optimization Target: Superior Overall Survival (OS)

13. Study Procedures & Methodology

Management Pathway: Second-line systemic oncology treatment
Education Protocol (HCP): Site initiation visits covering MK-2870 administration, RECIST 1.1 imaging criteria, and AE management.
Education Protocol (Patient): Informed consent process, IV infusion scheduling, and symptomatic AE reporting.
Quality Audit Mechanism: Blinded Independent Central Review (BICR) for impartial tumor imaging assessments.

14. Observation & Follow-Up Schedule

Total Duration: Approximately 4 years of trial involvement
Onsite Visit Cadence: Every 2 weeks (for MK-2870) or per TPC schedule (every 3 to 6 weeks)
Remote Monitoring Frequency: Regular AE tracking and survival follow-up
Checklist Review Interval: Scheduled tumor imaging assessments to evaluate RECIST 1.1 progression

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				